Minimal Surfaces

Fall 2026 — Scuola Normale Superiore, Pisa

Giada Franz

29 October 2026

These are minimalist transcriptions of the lecture notes of a 40 hours course I am currently teaching in Scuola Normale Superiore in Pisa. The transcription of the handwritten notes into LaTeX is currently partially automated using a LLM and revised by Andrea Rocca. Work in progress.

Contents

Introduction **3**
- 0.1 The Plateau problem and its history . . . 3
- 0.2 Why study minimal surfaces? . . . 4
- 0.3 Course outline . . . 5

1 Preliminaries in Riemannian geometry **7**
- 1.1 Smooth manifolds, tangent spaces, and vector fields . . . 7
- 1.2 Tensors and differential forms . . . 8
- 1.3 Riemannian metrics, distance, and volume . . . 8
- 1.4 The Levi–Civita connection . . . 9
- 1.5 Curvature . . . 10
- 1.6 Operators on manifolds . . . 11
- 1.7 Submanifolds . . . 11
- 1.8 Geodesics . . . 13
- 1.9 Exercises . . . 17

2 Classical theory for minimal surfaces **21**
- 2.1 The minimal surface equation . . . 21
- 2.2 Examples in $\mathbb{R}^3$ . . . 23
- 2.3 The minimal surface equation for graphs . . . 23
- 2.4 First consequences of minimality . . . 24
- 2.5 Monotonicity formula . . . 25
- 2.6 The halfspace theorem . . . 28
- 2.7 The Weierstrass representation . . . 29
- 2.8 The Bernstein theorem . . . 31
- 2.9 Exercises . . . 33

Introduction

This course aims to give an introduction to minimal surface theory, up to covering topics of current research. The content is inspired by many different references, main ones are the book by Colding–Minicozzi [7] and the notes by White [35] and Chodosh [5].

A classical motivating picture for studying minimal surfaces is a wire immersed in soapy water: after withdrawing it (and neglecting gravity), the soap film is expected to minimize its area among surfaces spanning the same boundary. Denote the wire by a smooth *Jordan curve* Γ (a simple closed curve) and a spanning surface by Σ, with $\partial\Sigma = \Gamma$.

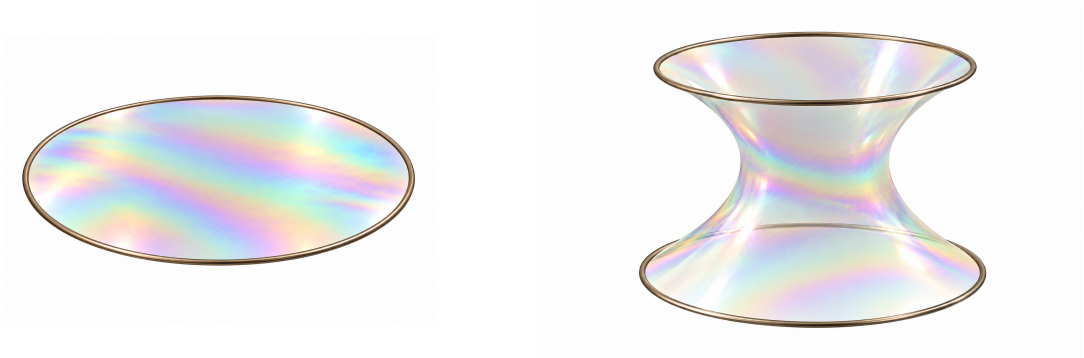


Figure 1: The Plateau problem: seek a least-area spanning surface Σ for a prescribed boundary curve Γ. A disk and a catenoid are familiar examples of minimal surfaces.

0.1 The Plateau problem and its history

Finding a surface with least area has a long history. In 1762, Lagrange [20] wrote down the equation for graphs of least area (the original memoir appeared in the volume dated 1760–1761). The problem of minimizing area with a fixed boundary curve Γ is named after Joseph Plateau, who investigated soap films (his findings were published in 1873 [29], after decades of research).

The classical Plateau problem was solved in the 1930s, independently by Jesse Douglas and Tibor Radó [10, 30]. Douglas received one of the first Fields Medals in 1936 for this work. An area-minimizing surface need not initially be given as an embedding: the construction may yield immersed surfaces and require an analysis of possible singularities and branch points.

For boundary curves in $\mathbb{R}^3$, work of Osserman and Gulliver in the 1970s ruled out interior branch points of area-minimizing disks [28, 16]. When $\Gamma \subset \partial K \subset \mathbb{R}^3$ for a compact

convex set K, Meeks–Yau [22] in 1982 established embeddedness results for least-area solutions.

Remark 0.1
It is an open problem whether branch points can occur at the (smooth) boundary.

Remark 0.2 - Branched minimal surfaces
The graph of the map $F\colon \mathbb{C} \to \mathbb{C}^2$ given by

$$F(z) = (z^2, z^3) \in \mathbb{C}^2$$

is a branched minimal surface.

Morrey [25] in 1948 generalized the problem to surfaces in Riemannian manifolds (M, g). There is also a higher-dimensional theory of area minimization. In the setting of geometric measure theory, the Federer–Fleming theory (1960) [13] provides least-mass representatives for appropriate integral homology classes, schematically

$$\sigma \in H_k(M; \mathbb{Z}) \quad \Longrightarrow \quad \text{an area-minimizing integral } k\text{-cycle representing } \sigma.$$

For codimension-one minimizers, write $\Sigma^n \subset M^{n+1}$. When $n \leq 6$ (ambient dimension at most 7), regularity theory yields smooth area-minimizing hypersurfaces in the interior, under the usual hypotheses; see also the foundational regularity work of Simons [33].

The subject subsequently developed in many directions: finding area-minimizing surfaces in new settings, constructing surfaces that are only critical points of the area functional (*minimal surfaces*), and understanding their geometry and topology.

0.2 Why study minimal surfaces?

Minimal surfaces are beautiful geometric objects in their own right. They also arise naturally in many problems as critical points or minimizers of geometric energies. Some examples are:

- **Scalar curvature and rigidity:** which Riemannian metrics on a torus T^n can have nonnegative scalar curvature? [31]
- **General relativity:** the positive mass theorem [32].
- **Knot theory:** a closed curve in $\mathbb{R}^3$ with total curvature less than 4π is unknotted [24]; counting minimal disks in hyperbolic 4-space $\mathbb{H}^4$ with prescribed knot as ideal boundary is related to knot invariants.
- **Image processing:** image denoising and optimal transport.
- **3-manifolds:** finding useful splittings, especially Heegaard splittings.
- **The Smale conjecture and diffeomorphism groups:** studying $\mathrm{Diff}(M)$ for a 3-manifold M.
- **Spectral theory.**

Some applications will be studied in the lectures; others are proposed as possible exam topics.

0.3 Course outline

1. Preliminaries in Riemannian geometry.
2. Foundations of minimal surface theory, including minimal surfaces in Euclidean space.
3. Minimal surfaces in positive curvature.
4. Minimal surfaces in negative curvature.
5. Existence of minimal surfaces.

Chapter 1

Preliminaries in Riemannian geometry

1.1 Smooth manifolds, tangent spaces, and vector fields

Let M^n be a smooth manifold and $p \in M$. Its tangent space T_pM may be defined as the vector space of velocities $\gamma'(0)$ of smooth curves γ passing through p. Equivalently, it consists of derivations at p, i.e., linear maps

$$v\colon C^\infty(M) \longrightarrow \mathbb{R}, \qquad v(fg) = f(p)v(g) + g(p)v(f).$$

Under this identification, the tangent vector $v = \gamma'(0)$ acts by

$$\gamma'(0)(f) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} f(\gamma(t)).$$

The *tangent bundle* is the disjoint union

$$TM = \bigsqcup_{p\in M} T_pM.$$

A chart $(x^1, \dots, x^n)\colon U \subset M \to \mathbb{R}^n$ induces a basis $\left\{\partial_i|_p = \frac{\partial}{\partial x^i}\Big|_p\right\}_{i=1}^n$ of T_pM.

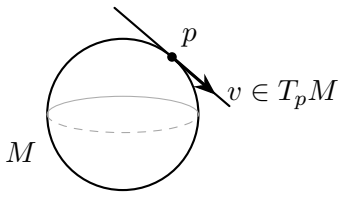


Figure 1.1: Schematic view of a tangent vector at a point of a manifold.

For a smooth map $F : M \to N$ between smooth manifolds, the *differential* at p is

$$\mathrm{d}F_p\colon T_pM \longrightarrow T_{F(p)}N, \qquad (\mathrm{d}F_p(v))(f) = v(f \circ F).$$

It satisfies the chain rule

$$\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p.$$

The *cotangent space* is the dual $T_p^*M = (T_pM)^*$, and the *cotangent bundle* is T^*M. For $f \in C^\infty(M)$,

$$\mathrm{d}f_p \in T_p^*M, \qquad \mathrm{d}f_p(v) = v(f).$$

In local coordinates, the covectors $\mathrm{d}x^i|_p$ form the dual basis to the $\partial_i|_p$ and

$$\mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i.$$

Here and below, repeated upper and lower indices may be summed over, in Einstein notation.

A *smooth vector field* $X \in \mathfrak{X}(M) = \Gamma(TM)$ is a smooth section of the tangent bundle. Locally,

$$X = X^i\partial_i, \qquad X(f) = \mathrm{d}f(X).$$

Every smooth vector field has a local *flow* Φ_t^X, defined by

$$\frac{\partial}{\partial t}\Phi_t^X(p) = X(\Phi_t^X(p)), \qquad \Phi_0^X(p) = p.$$

If M is compact, or more generally if X has compact support, the flow is defined for all $t \in \mathbb{R}$, and each $\Phi_t^X \colon M \to M$ is a diffeomorphism.

1.2 Tensors and differential forms

An (r, s)-tensor field is a $C^\infty(M)$-multilinear map

$$T\colon \underbrace{\Gamma(T^*M) \times \cdots \times \Gamma(T^*M)}_{r \text{ factors}} \times \underbrace{\Gamma(TM) \times \cdots \times \Gamma(TM)}_{s \text{ factors}} \longrightarrow C^\infty(M).$$

We denote by $T^{(r,s)}(M)$ the space of (r, s)-tensor fields. In coordinates,

$$T = T^{i_1\cdots i_r}_{j_1\cdots j_s}\,\partial_{i_1} \otimes \cdots \otimes \partial_{i_r} \otimes \mathrm{d}x^{j_1} \otimes \cdots \otimes \mathrm{d}x^{j_s}.$$

In particular, vector fields are $(1, 0)$-tensor fields.

A *k-form* $\omega \in \Omega^k(M)$ is an alternating $(0, k)$-tensor. Thus

$$\omega(\ldots, X_i, \ldots, X_j, \ldots) = -\omega(\ldots, X_j, \ldots, X_i, \ldots).$$

The *wedge product* is the alternating tensor product: for $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$,

$$(\alpha \wedge \beta)(X_1, \ldots, X_{k+\ell}) = \frac{1}{k!\ell!}\sum_{\sigma \in S_{k+\ell}} \operatorname{sgn}(\sigma)\alpha(X_{\sigma(1)}, \ldots, X_{\sigma(k)})\beta(X_{\sigma(k+1)}, \ldots, X_{\sigma(k+\ell)}).$$

A manifold of dimension n is *orientable* if it admits a nowhere-vanishing smooth n-form.

1.3 Riemannian metrics, distance, and volume

A *Riemannian manifold* (M^n, g) is a smooth manifold endowed with a smooth, symmetric, positive-definite $(0, 2)$-tensor g, its Riemannian metric. It defines an inner product $\langle \cdot, \cdot \rangle_p$ on each T_pM. Locally,

$$g = g_{ij}\,\mathrm{d}x^i \otimes \mathrm{d}x^j, \qquad (g^{ij}) = (g_{ij})^{-1}.$$

The *Riemannian distance* is the infimum of lengths of piecewise smooth curves joining two points:

$$d(p,q) = \inf_{\substack{\gamma\colon [a,b]\to M \\ \gamma(a)=p,\,\gamma(b)=q}} \ell(\gamma), \qquad \ell(\gamma) = \int_a^b |\gamma'(t)|_g \mathrm{d}t.$$

If M is oriented, its *Riemannian volume form* $\mathrm{d}\,\mathrm{vol}_g$ takes value 1 on each positively oriented orthonormal basis $e_1, \ldots, e_n$. In oriented coordinates,

$$\mathrm{d}\,\mathrm{vol}_g = \sqrt{\det(g_{ij})}\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n,$$

so integration becomes

$$\int_M f\,\mathrm{d}\,\mathrm{vol}_g = \int_{x(U)} (f \circ x^{-1}) \sqrt{\det(g_{ij})}\,\mathrm{d}x^1 \cdots \mathrm{d}x^n$$

when f is supported inside a positively oriented coordinate chart U.

Theorem 1.1 - Stokes' theorem
For a smooth oriented n-manifold M (possibly with boundary) and a compactly supported $(n-1)$-form ω,

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega,$$

using the induced orientation on ∂M.

1.4 The Levi–Civita connection

A *connection* on the tangent bundle assigns to vector fields X, Y a new vector field $\nabla_X Y$. The *Levi–Civita connection* is characterized by the following conditions:

(i) ∇ is $C^\infty(M)$-linear in X, $\mathbb{R}$-linear in Y, and satisfies the Leibniz rule

$$\nabla_X(fY) = X(f)Y + f\nabla_X Y.$$

(ii) *Torsion-free:*

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

(iii) *Metric compatible:*

$$X\big(g(Y,Z)\big) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

In local coordinates the *Christoffel symbols* Γ_{ij}^k are defined by

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

On Euclidean space $\mathbb{R}^n$, the Levi–Civita connection is the ordinary directional derivative:

$$\nabla_X^{\mathbb{R}^n} Y = D_X Y = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} Y(p + tX(p)).$$

Notation 1.1
Along a curve $\gamma\colon [a,b] \to M$, covariant differentiation of a vector field Y along γ is written

$$\frac{\nabla Y}{\mathrm{d}t} = \nabla_{\gamma'(t)} Y.$$

For a vector field Y *along* γ we abbreviate its first and second covariant derivatives by

$$Y' := \frac{\nabla Y}{\mathrm{d}t}, \qquad Y'' := \frac{\nabla}{\mathrm{d}t} Y'.$$

Thus a prime on a vector field along γ means *covariant* differentiation; in $\mathbb{R}^n$ this agrees with the usual derivative. The connection also extends, via the Leibniz rule, to tensor fields.

1.5 Curvature

The *Riemann curvature tensor* R of (M^n, g) is defined by

$$R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

We also use the associated $(0,4)$-tensor, with the following convention for the last two arguments:

$$R(X,Y,Z,W) := g(R(X,Y)W, Z).$$

The *sectional curvature* of the plane spanned by linearly independent vectors $X, Y \in T_pM$ is

$$\mathrm{Sec}(X,Y) = \frac{R(X,Y,X,Y)}{|X|^2|Y|^2 - g(X,Y)^2}.$$

The *Ricci curvature* is the trace

$$\mathrm{Ric}(X,Y) = \sum_{i=1}^{n} R(X, e_i, Y, e_i),$$

where $(e_i)_{i=1}^n$ is an orthonormal basis of T_pM. The *scalar curvature* is

$$\mathrm{Scal} = \mathrm{tr}_g \,\mathrm{Ric} = \sum_{i \neq j} \mathrm{Sec}(e_i, e_j).$$

The curvature tensor satisfies the following symmetries and identities:

$$\begin{aligned} R(X,Y,Z,W) &= R(Z,W,X,Y) = -R(X,Y,W,Z), \\ R(X,Y)Z + R(Y,Z)X + R(Z,X)Y &= 0 \qquad \text{(first Bianchi identity)}, \\ (\nabla_X R)(Y,Z) + (\nabla_Y R)(Z,X) + (\nabla_Z R)(X,Y) &= 0 \qquad \text{(second Bianchi identity)}. \end{aligned}$$

In the last identity, R is viewed as a $(1,3)$-tensor and the equality holds after applying both sides to a fourth vector.

Example 1.2 - Space forms
The model spaces $\mathbb{R}^n$, S^n (round unit sphere), and $\mathbb{H}^n$ have constant sectional curvature $\kappa = 0, 1, -1$, respectively. More generally, on a manifold of constant sectional curvature κ,

$$\mathrm{Ric} = (n-1)\kappa\, g, \qquad \mathrm{Scal} = n(n-1)\kappa.$$

These models are, up to isometry, the complete simply connected space forms in their respective curvature classes.

1.6 Operators on manifolds

The metric induces the *musical isomorphisms*

$$\begin{aligned} \flat\colon TM &\longrightarrow T^*M, \qquad X^\flat(Y) = g(X,Y), \\ \sharp\colon T^*M &\longrightarrow TM, \qquad g(\omega^\sharp, Y) = \omega(Y). \end{aligned}$$

In coordinates,

$$(X^\flat)_j = g_{ij}X^i, \qquad (\omega^\sharp)^i = g^{ij}\omega_j.$$

For $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$, we define

$$\begin{aligned} \nabla f &:= (\mathrm{d}f)^\sharp && \text{(gradient)}, \\ \operatorname{div} X &:= \operatorname{tr}(\nabla X) = \sum_{i=1}^n g(\nabla_{e_i} X, e_i) && \text{(divergence)}, \\ \nabla^2 f &:= \nabla(\mathrm{d}f), \quad \nabla^2 f(X,Y) = g(\nabla_X \nabla f, Y) && \text{(Hessian)}, \\ \Delta f &:= \operatorname{div}(\nabla f) = \operatorname{tr}_g(\nabla^2 f) && \text{(Laplacian)}. \end{aligned}$$

Here (e_i) is a local orthonormal frame. In particular, the convention for Δ is $\Delta = \operatorname{div}\nabla$.

Theorem 1.2 - Divergence theorem
Let (M^n, g) be oriented, with smooth boundary, and let X be a compactly supported smooth vector field (smooth up to the boundary). If ν is the outward-pointing unit normal to ∂M, then

$$\int_M \operatorname{div} X \,\mathrm{d}\operatorname{vol}_g = \int_{\partial M} g(X,\nu)\,\mathrm{d}\operatorname{vol}_{\partial M}.$$

Proof. By inserire Cartan's formula for the Lie derivative and the fact that a top-degree form is closed,

$$\mathrm{d}(\iota_X \mathrm{d}\operatorname{vol}_g) = \mathcal{L}_X \mathrm{d}\operatorname{vol}_g = (\operatorname{div} X)\,\mathrm{d}\operatorname{vol}_g.$$

Moreover, on ∂M one has

$$(\iota_X \mathrm{d}\operatorname{vol}_g)|_{\partial M} = g(X,\nu)\,\mathrm{d}\operatorname{vol}_{\partial M}.$$

The assertion follows from Stokes' theorem. □

Remark 1.3
An alternative proof in local coordinates, using the divergence formula and Gauss's theorem in $\mathbb{R}^n$, is left as an exercise.

1.7 Submanifolds

We use $\Sigma^k \subset M^n$ for a k-dimensional submanifold of an n-dimensional ambient manifold. For hypersurfaces we instead write $\Sigma^n \subset M^{n+1}$.

For a k-dimensional submanifold S^k, we use $\operatorname{vol}(S)$ and $|S|$ interchangeably for its k-dimensional Riemannian volume (measure). Thus $|\Sigma|$ denotes k-dimensional volume when $\Sigma^k \subset M^n$, whereas $|\partial\Sigma|$ denotes the $(k-1)$-dimensional measure of its boundary.

For a smooth map $F\colon \Sigma^k \to M^n$:

- F is an *immersion* if $\mathrm{d}F_p$ is injective for every $p \in \Sigma$;

- F is an *embedding* if it is an immersion and a homeomorphism onto its image (with the subspace topology);
- F is an *isometry* if $F^* g_M = g_\Sigma$, where g_Σ, g_M are metrics on Σ, M.

From now on, unless otherwise stated, we assume that $F\colon \Sigma^k \to M^n$ is an isometric immersion. As usual, we identify Σ with its image locally and regard tangent vector fields as fields along the immersion. The intrinsic Levi–Civita connection is the tangential projection of the ambient one:

$$\nabla^\Sigma_X Y = (\nabla^M_X Y)^\top.$$

Notation 1.4
In this subsection, for brevity, write $\nabla := \nabla^\Sigma$ and $D := \nabla^M$.

Definition 1.3 - Second fundamental form
The *second fundamental form* is the normal-bundle-valued symmetric bilinear form

$$A(X,Y) := (D_X Y)^\perp \in T\Sigma^\perp.$$

For tangent fields X, Y, it satisfies

$$A(X,Y) = A(Y,X), \qquad D_X Y = \nabla_X Y + A(X,Y).$$

The (unnormalized) *mean curvature vector* is

$$\vec{H} := \operatorname{tr}_{T\Sigma} A = \sum_{i=1}^{k} A(e_i, e_i).$$

Proposition 1.4 - Gauss equation
For tangent vectors $X, Y, Z, W \in T_p\Sigma$,

$$R^\Sigma(X,Y,Z,W) = R^M(X,Y,Z,W) + g(A(X,Z),A(Y,W)) - g(A(X,W),A(Y,Z)).$$

Proposition 1.5 - Codazzi equation
For tangent fields X, Y, Z,

$$(\nabla_X A)(Y,Z) - (\nabla_Y A)(X,Z) = \big(R^M(X,Y)Z\big)^\perp,$$

where ∇A uses the induced tangent and normal connections.

Hypersurfaces

For a hypersurface $\Sigma^n \subset M^{n+1}$, we say Σ is *two-sided* if there exists a global unit normal field $N\colon \Sigma \to T\Sigma^\perp$. Once N is chosen, define

$$\mathrm{II}(X,Y) := -g(A(X,Y),N), \qquad H := -g(\vec{H}, N).$$

Thus $H = \operatorname{tr}_{T\Sigma} \mathrm{II}$.

Example 1.5 - Round sphere
For $S^n \subset \mathbb{R}^{n+1}$ with the outward unit normal $N(x) = x$,

$$A(X,Y) = -g(X,Y)N, \qquad \mathrm{II}(X,Y) = g(X,Y),$$

and hence $\vec{H} = -nN$ and $H = n$.

Definition 1.6 - Shape operator
For a chosen unit normal N, the *shape operator* is

$$S_N \colon T_p\Sigma \longrightarrow T_p\Sigma, \qquad S_N(X) := -(D_X N)^\top.$$

It satisfies

$$g(S_N(X), Y) = g(A(X,Y), N) = -\,\mathrm{II}(X,Y).$$

The *principal curvatures* (with this convention) are the eigenvalues of S_N.

Remark 1.6
With the signs above, the eigenvalues of S_N are the negatives of those of the quadratic form II with respect to g. In particular, $S_N = -\,\mathrm{id}$ for the outward-oriented unit sphere.

1.8 Geodesics

Let (M^n, g) be a complete Riemannian manifold (for example, $S^3 \subset \mathbb{R}^4$ with the round metric). Given $p, q \in M$, what is the shortest path between them?

For a piecewise smooth curve $\gamma \colon I = [a,b] \to M$, define its *length* by

$$\ell(\gamma) = \int_a^b |\gamma'(t)|\mathrm{d}t, \qquad |\gamma'(t)| = \sqrt{g(\gamma'(t), \gamma'(t))}.$$

Given two points p, q, we seek a curve γ between them minimizing $\ell(\gamma)$.

Rather then minimizing directly the length functional, it is often convenient to minimize the *energy*

$$E(\gamma) = \frac{1}{2}\int_a^b |\gamma'(t)|^2 \mathrm{d}t.$$

Indeed, by Cauchy-Schwarz, one has:

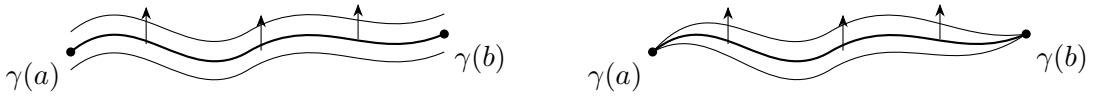
$$\ell(\gamma)^2 \leq 2(b-a)E(\gamma),$$

with equality if and only if γ has constant speed. Since a repametrization of a curve γ does *not* change its image, we can always find a diffeomorphism $\sigma \colon I \to I$ so that $\gamma(\sigma(t))$ has constant speed. Thus energy minimizers correspond to length minimizers parametrized with constant speed. One of the main advantages of working with E instead of ℓ is its better analytical structure, which makes it particularly suitable for proving the existence of minimizers.

Definition 1.7
A smooth map $F \colon I \times (-\varepsilon, \varepsilon) \to M$ is a *variation* of γ if $F(t,0) = \gamma(t)$. It is *proper* (fixed endpoints) if $F(a,s) = \gamma(a)$ and $F(b,s) = \gamma(b)$ for every s. Its associated

variational field is

$$X(t) = \left.\frac{\partial F(t,s)}{\partial s}\right|_{s=0}.$$



Variation with moving endpoints — Proper variation (fixed endpoints)

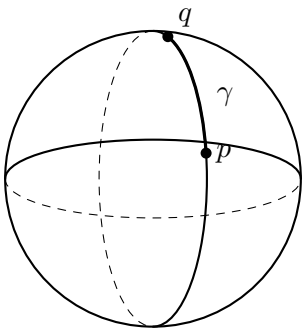
Figure 1.2: Two variations of a curve. On the left the endpoints move; on the right they remain fixed (a proper variation).

Proposition 1.8 - First variation of energy
For a variation F of γ with variational field X,

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}E(F(\cdot,s))\right|_{s=0} = -\int_a^b \left\langle X, \frac{\nabla}{\mathrm{d}t}\gamma' \right\rangle \mathrm{d}t + \langle X(b), \gamma'(b)\rangle - \langle X(a), \gamma'(a)\rangle.$$

Definition 1.9
A smooth curve $\gamma\colon I \to M$ is a *geodesic* if $\frac{\nabla}{\mathrm{d}t}\gamma' = 0$ on I. Equivalently, it is a critical point of the energy with respect to proper variations. A *(minimizing) geodesic segment* is a geodesic which minimizes the length between its endpoints.

Example 1.7 - Geodesics on S^2
Great circles are geodesics on the round sphere. A long arc of a great circle need not be length minimizing between its endpoints.



γ is a geodesic segment: an arc of a great circle on the round sphere.

Figure 1.3: The round sphere S^2 with a geodesic γ joining two points p and q. Great circles are exactly the geodesics of the round metric.

Remark 1.8
Every minimizing segment is a geodesic, but the converse need not hold globally. Every geodesic is locally a minimizing segment.

Proposition 1.10 - Local existence and uniqueness
For every $p \in M$, there is a neighborhood U_p of $0 \in T_pM$ such that for any $v \in U_p$, there is a unique geodesic starting at p with initial velocity v, defined at least until time 1. The exponential map is $\exp_p(v) = \gamma_v(1)$.

Remark 1.9
The previous result is essentially the local existence for solutions to a sistem of ODEs. What can be said about the *global* existence?

Definition 1.11
Let M be a manifold. A connection ∇ on M is *geodesically complete* if the exponential map $\exp_p$ is defined on the full tangent space at $p \in M$. Equivalently, every maximal geodesic $\gamma(t)$ in M needs to be defined for all times $t \in \mathbb{R}$. A Riemannian manifold M is *geodesically complete* if its Levi-Civita connection is.

Theorem 1.12 - Hopf-Rinow
Let M be a connected Riemannian manifold. The following are equivalent:

1. *M is geodesically complete,*
2. *Given a subset $K \subset M$ then*

$$K \text{ compact} \iff K \text{ closed and bounded,}$$

3. *M is complete.*

Proposition 1.13 - Second variation of energy
Suppose γ is a geodesic and F is a variation of γ with variational field X. With the curvature convention used in these notes,

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}s^2}E(F(\cdot,s))\bigg|_{s=0} &= \int_a^b (|X'|^2 - \langle R(X,\gamma')\gamma', X\rangle)\mathrm{d}t + \left[\langle \nabla_s \partial_s F|_{s=0}, \gamma'\rangle\right]_a^b \\ &= -\int_a^b \langle X, X'' + R(X,\gamma')\gamma'\rangle \,\mathrm{d}t + \langle X, X'\rangle|_a^b + \left[\langle \nabla_s \partial_s F|_{s=0}, \gamma'\rangle\right]_a^b .\end{aligned}$$

For a proper variation

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}E(F(\cdot,s))\bigg|_{s=0} = -\int_a^b \langle X, X'' + R(X,\gamma')\gamma'\rangle \,\mathrm{d}t. \tag{1.1}$$

both boundary terms vanish.

Definition 1.14
The *Jacobi operator* along a geodesic is

$$J(X) = X'' + R(X,\gamma')\gamma'.$$

A field in the kernel of J is called a *Jacobi field*.

Theorem 1.15 - Myers [26]
Let (M^n, g) be complete and suppose

$$\mathrm{Ric} \geq (n-1)\kappa\, g \qquad \text{for some } \kappa > 0.$$

Then

1. *M is compact and*
$$\mathrm{diam}(M) \leq \frac{\pi}{\sqrt{\kappa}};$$
2. *$\pi_1(M)$ is finite.*

Idea. If we are able to construct a proper variation F then thanks to 1.1 and the bound on the curvature we could infer the desired result. Having only a bound on Ric and not on $R(\cdot,\cdot)$, we need to "average". ⌟

Proof. For (i), let $\gamma\colon [0,\ell] \to M$ be a unit-speed geodesic. It suffices to show that if $\ell > \pi/\sqrt{\kappa}$, then $\gamma|_{[0,\ell]}$ is not minimizing.

Choose an orthonormal basis $E_\alpha(0)$, $\alpha = 1,\dots,n-1$, of $\gamma'(0)^\perp \subset T_pM$, and parallel-transport[1] it along γ; thus $E'_\alpha = 0$. Set $c = (\pi/\ell)^2 < \kappa$ and define

$$\mathrm{sn}_c(t) = \frac{1}{\sqrt{c}}\sin(\sqrt{c}\,t), \qquad V_\alpha(t) = \mathrm{sn}_c(t)E_\alpha(t), \qquad \mathrm{sn}_c'' + c\,\mathrm{sn}_c = 0.$$

The proper variations $F_\alpha(t,s) = \exp_{\gamma(t)}(sV_\alpha(t))$ have second variations whose sum equals

$$\begin{aligned}
\sum_{\alpha=1}^{n-1} \frac{\mathrm{d}^2}{\mathrm{d}s^2} E(F_\alpha(\cdot,s))\Big|_{s=0} &= \sum_{\alpha=1}^{n-1} \int_0^\ell \Big(|V_\alpha'|^2 - \langle R(V_\alpha,\gamma')\gamma', V_\alpha\rangle \Big)\,\mathrm{d}t \\
&= \int_0^\ell \Big((n-1)(\mathrm{sn}_c')^2 - \mathrm{sn}_c^2\,\mathrm{Ric}(\gamma',\gamma') \Big)\,\mathrm{d}t \\
&\leq (n-1)\int_0^\ell \big((\mathrm{sn}_c')^2 - \kappa\,\mathrm{sn}_c^2 \big)\mathrm{d}t \\
&= (n-1)\int_0^\ell (c-\kappa)\,\mathrm{sn}_c^2\,\mathrm{d}t < 0.
\end{aligned}$$

Here integration by parts and $\mathrm{sn}_c(0) = \mathrm{sn}_c(\ell) = 0$ give $\int(\mathrm{sn}_c')^2 = c\int \mathrm{sn}_c^2$. Thus at least one proper variation has negative second variation, contradicting minimizing behavior. Completeness and Hopf–Rinow now imply compactness and the diameter bound.

For (ii), let $\pi : \widetilde{M} \to M$ be the universal cover equipped with the pullback metric $\widetilde{g} = \pi^* g$. The cover is complete and satisfies $\widetilde{\mathrm{Ric}} \geq (n-1)\kappa\widetilde{g}$. By (i), $\widetilde{M}$ is compact. A compact universal cover has only finitely many sheets over M, so $\pi_1(M)$ is finite. □

Remark 1.10
With this result we have a nice example of a possible use of the theory of minimal surfaces on a manifold. The theorem uses geodesics, i.e. one dimensional minimal surfaces, to infer global geometric and topological information about a manifold.

[1] Namely, $E'_\alpha(t) = \frac{\nabla}{dt}E_\alpha = 0$. Writing it down in coordinates will actually give a ODE.

1.9 Exercises

Exercise 1.11 - Mean curvature of graphs
Let $f\colon \Omega \to \mathbb{R}$ be a smooth function with $\Omega \subset \mathbb{R}^n$ smooth open set. Let $\Sigma = \operatorname{graph} f \subset \mathbb{R}^{n+1}$ be the graph of f with metric tensor g inherited by $\mathbb{R}^{n+1}$.

(i) Show that Σ is a hypersurface in $\mathbb{R}^{n+1}$, by finding a global chart (Ω, φ) for Σ.

(ii) Denoting by ∂_i and ∇ the usual differentiation on $\mathbb{R}^n$, by N the upward unit normal field of Σ, and by II_N and H the second fundamental form and the mean curvature of Σ in direction N respectively. Prove that

$$g_{ij} = \delta_{ij} + \partial_i f \partial_j f, \qquad N = \frac{(-\nabla f, 1)}{\sqrt{1+|\nabla f|^2}}, \qquad \mathrm{II}_N = \frac{\nabla^2 f}{\sqrt{1+|\nabla f|^2}},$$

$$H = \operatorname{div}\left(\frac{\nabla f}{\sqrt{1+|\nabla f|^2}}\right) = \frac{\Delta f}{\sqrt{1+|\nabla f|^2}} - \frac{(\nabla^2 f)(\nabla f, \nabla f)}{\left(\sqrt{1+|\nabla f|^2}\right)^3}.$$

Exercise 1.12 - Two-sidedness of hypersurfaces
Show that every properly embedded hypersurface Σ^n in $\mathbb{R}^{n+1}$ is two-sided.

Exercise 1.13 - Divergence and Laplacian
Let (M^n, g) be a Riemannian manifold.

(i) Prove the expression of the divergence with respect to the metric in coordinates

$$\operatorname{div}(X) = \frac{1}{\sqrt{\det(g_{ij})}} \partial_k \left(\sqrt{\det(g_{ij})} X^k\right).$$

(ii) Prove the expression of the Laplacian with respect to the metric in coordinates

$$\Delta f = \frac{1}{\sqrt{\det(g_{ij})}} \partial_k \left(\sqrt{\det(g_{ij})} g^{kl} \partial_l f\right).$$

(iii) Show that, if $f \in C_c^\infty(M)$ and $\Delta f \geq 0$, then f is constant.

Exercise 1.14 - Normal frame, normal chart, exponential normal chart
Let us fix a Riemannian manifold (M^n, g) and a point $p \in M$. Let us set $\{(e_i)|_p\}_{i=1}^n$ an orthonormal basis of T_pM.

(i) Show that there exists a local orthonormal frame $\{e_i\}_{i=1}^n$ of TM around p such that $e_i(p) = (e_i)|_p$, and $(\nabla_{e_i} e_j)(p) = 0$ for every $1 \leq i, j \leq n$. Notice that then, in such a frame, the Christoffel symbols are zero at p, i.e., $\Gamma_{ij}^k(p) = 0$ for every $1 \leq i, j, k \leq n$. We will call such a frame a *normal frame* around p.

(ii) Show that there exists a chart (U, φ) around p such that $\partial_i(p) = (e_i)|_p$ and $(\nabla \partial_i)(p) = 0$ for every $1 \leq i \leq n$. Show that if in this chart we have $g = g_{ij} dx^i \otimes dx^j$, then $g_{ij}(p) = \delta_{ij}(p)$, $\partial_k g_{ij}(p) = 0$, and $\Gamma_{ij}^k(p) = 0$ for every $1 \leq i, j, k \leq n$. We will call such a chart a *normal chart* around p. Notice that one example of such a chart is the *exponential normal chart*, that gives rise to *exponential normal coordinates*.

Exercise 1.15 - Warped products $I \times_f N$
Let (N^n, g) be a Riemannian manifold and let $I \subset \mathbb{R}$ be an interval. Given $f \in C^\infty(I)$ with $f \neq 0$ on I, we define the *warped product* manifold $M = I \times_f N$ to be the $(n+1)$-dimensional manifold $I \times N$ endowed with the warped metric

$$\overline{g} = dt^2 + f^2(t)g.$$

Let $e_1, \dots, e_n$ be a local orthonormal frame on an open set $U \subset N$. Assume also that $\{e_i\}$ is normal at some $x \in U$. Let also

$$\overline{e}_0 = \partial_t, \qquad \overline{e}_i = \frac{1}{f} e_i.$$

(i) Observe that $\overline{e}_0, \overline{e}_1, \dots, \overline{e}_n$ is an orthonormal frame on $I \times U$.

(ii) If R and $\overline{R}$ are the Riemann tensor on N and $I \times_f N$ respectively, prove that

$$\overline{R}_{0ijk} = \overline{R}_{i0jk} = 0, \qquad \overline{R}_{i0j0} = -\frac{f''}{f}\delta_{ij}, \qquad \overline{R}_{ijkl} = \frac{1}{f^2}R_{ijkl} - \left(\frac{f'}{f}\right)^2 (\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}),$$

for any $i, j, k, l = 1, \dots, n$.

(iii) If N has constant sectional curvature $c \in \mathbb{R}$, deduce that

$$\overline{\sec}(\nabla t \wedge \overline{e}_i) = -\frac{f''}{f}, \qquad \overline{\sec}(\overline{e}_i \wedge \overline{e}_j) = \frac{c - (f')^2}{f^2}$$

for any $i, j = 1, \dots, n$ with $i \neq j$.

Exercise 1.16 - Radial models of constant sectional curvature
A *radial model of constant sectional curvature* $c \in \mathbb{R}$ is

$$\left((0, T) \times \mathbb{S}^{n-1}, \langle \cdot, \cdot \rangle := \mathrm{d}r^2 + \mathrm{sn}_c(r)^2 \mathrm{d}\theta^2\right),$$

where $\mathbb{S}^{n-1}$ is the $(n-1)$-dimensional sphere of radius one and $\mathrm{d}\theta^2$ is the standard metric on it; the function $\mathrm{sn}_c(r)$ is the positive solution of the Jacobi equation

$$\begin{cases} \mathrm{sn}_c''(r) + c\,\mathrm{sn}_c(r) = 0, \\ \mathrm{sn}_c(0) = 0, \\ \mathrm{sn}_c'(0) = 1; \end{cases}$$

and $T := \sup\{\varrho \ : \ \mathrm{sn}_c(r) > 0 \ \forall r \in (0, \varrho)\}$.

(i) Prove that

$$\mathrm{sn}_c(r) = \begin{cases} \frac{1}{\sqrt{c}} \sin(\sqrt{c}\, r) & \text{if} \quad c > 0, \quad \text{then} \quad T = \frac{\pi}{\sqrt{c}}, \\ r & \text{if} \quad c = 0, \quad \text{then} \quad T = +\infty, \\ \frac{1}{\sqrt{-c}} \sinh(\sqrt{-c}\, r) & \text{if} \quad c < 0, \quad \text{then} \quad T = +\infty. \end{cases}$$

(ii) Show that, given $r(\cdot) := d(0, \cdot)$ the distance from the origin in this model, we have

$$\nabla^2 r(\cdot, \cdot) = \frac{\mathrm{sn}_c'(r)}{\mathrm{sn}_c(r)} \left(\langle \cdot, \cdot \rangle - \mathrm{d}r \otimes \mathrm{d}r(\,\cdot, \cdot)\right).$$

(iii) Show that these models have actually constant sectional curvature c.

Exercise 1.17 - The round sphere
For $n \geq 2$, let $\mathbb{S}^n_r \coloneqq \{x \in \mathbb{R}^{n+1} \ : \ |x| = r\}$ be the round sphere of radius r, with the metric g inherited by $\mathbb{R}^{n+1}$ and the outward unit normal $N(x) = x/r$.

(i) Show that $\mathbb{S}^n_r$ has second fundamental form $A(X,Y) = -\frac{1}{r}\langle X, Y\rangle N$, second fundamental form in the normal direction $\mathrm{II}(X,Y) = \frac{1}{r}\langle X, Y\rangle$, and shape operator $S_N = -\frac{1}{r}\mathrm{id}$. Deduce that the vector and scalar mean curvatures of $\mathbb{S}^n_r$ are equal to $\vec{H} = -\frac{n}{r}N$ and $H = \frac{n}{r}$.

(ii) Show that the curvature tensor of $\mathbb{S}^n_r$ is

$$R(X,Y,Z,W) = \frac{1}{r^2}\big(\langle X, Z\rangle\langle Y, W\rangle - \langle X, W\rangle\langle Y, Z\rangle\big).$$

Deduce that $\mathbb{S}^n_r$ has constant sectional curvature $\frac{1}{r^2}$, Ricci curvature $\mathrm{Ric} = \frac{n-1}{r^2}\, g$, and scalar curvature $\mathrm{Scal} = \frac{n(n-1)}{r^2}$.

Chapter 2

Classical theory for minimal surfaces

2.1 The minimal surface equation

Let (M^n, g) be a complete Riemannian manifold and $\Sigma^k \subset M^n$ an embedded[1] submanifold.

Definition 2.1
For a smooth vector field X along Σ, define its divergence along Σ by

$$\operatorname{div}_\Sigma X = \sum_{i=1}^{k} g(\nabla_{e_i} X, e_i) = \operatorname{tr}_{T\Sigma}(\nabla X),$$

where $\{e_1, \dots, e_k\}$ is an orthonormal basis of $T_p\Sigma$.

Remark 2.1 - Tangential vs. ambient divergence
Consider $\Sigma^k \subset M^n$ as above. Suppose that X is a smooth vector field defined on an open neighborhood of Σ in M. If $\{e_1, \dots, e_k\}$ is an orthonormal basis of $T_p\Sigma$, completed to an orthonormal basis $\{e_1, \dots, e_n\}$ of T_pM, then

$$\operatorname{div}_M X = \operatorname{div}_\Sigma X + \sum_{\alpha=k+1}^{n} g(\nabla_{e_\alpha} X, e_\alpha).$$

In particular, the tangential divergence only involves derivatives along directions tangent to Σ. Moreover, if X is defined only along Σ, we can compute $\operatorname{div}_\Sigma X$ using any smooth extension of X to a neighborhood in M. The result is independent of the chosen extension.

Achtung: In general, $\operatorname{div}_\Sigma X$ does not coincide with the ambient divergence $\operatorname{div}_M X$, since only tangential derivatives are taken into account.

In particular, if Σ is a hypersurface with unit normal ν, then

$$\boxed{\operatorname{div}_\Sigma X = \operatorname{div}_M X - g(\nabla_\nu X, \nu).}$$

[1] Much of the discussion also applies to immersed submanifolds. We consider embedded for sake of simplicity.

Notice that X need not be tangent to Σ. However, if X is tangent to Σ, its tangential divergence coincides with the intrinsic divergence associated with the induced metric.

Definition 2.2
A *compactly supported variation* of Σ is a smooth map $F\colon \Sigma \times (-\varepsilon, \varepsilon) \to M$ such that $F(x,0) = x$ and $F(x,t) = x$ outside a compact subset of Σ. Its variational field is

$$X(x) = \left.\frac{\partial F(x,t)}{\partial t}\right|_{t=0}.$$

Proposition 2.3 - First variation of volume
The first variation of the k-dimensional volume of Σ is

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathrm{vol}(F_t(\Sigma)) = \int_\Sigma \mathrm{div}_\Sigma X \,\mathrm{d}\,\mathrm{vol}_\Sigma \tag{2.1}$$

$$= -\int_\Sigma \langle \vec{H}, X\rangle \mathrm{d}\,\mathrm{vol}_\Sigma + \int_{\partial\Sigma} \langle X, \eta\rangle \mathrm{d}\,\mathrm{vol}_{\partial\Sigma}, \tag{2.2}$$

where $\vec{H}$ is the (unnormalized) mean curvature vector and η is the outward pointing unit conormal along $\partial\Sigma$.

Notation 2.2
We will write

$$\delta\Sigma(X) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathrm{vol}(F_t(\Sigma)).$$

Proof. Let us start with proving (2.1). Choose local coordinates $x^1, \dots, x^k$ on Σ and write

$$g_{ij}(t) = g(\partial_i F, \partial_j F), \qquad \mathrm{d}\,\mathrm{vol}_{F_t(\Sigma)} = \sqrt{\det g_{ij}(t)}\mathrm{d}x^1 \cdots \mathrm{d}x^k.$$

The ratio of induced volume densities is

$$v(t) := \frac{\mathrm{d}\,\mathrm{vol}_{F_t(\Sigma)}}{\mathrm{d}\,\mathrm{vol}_\Sigma} = \frac{\sqrt{\det g_{ij}(t)}}{\sqrt{\det g_{ij}(0)}}.$$

At a point $p \in \Sigma$ choose coordinates orthonormal at $t = 0$, so $g_{ij}(0) = \delta_{ij}$. Then

$$\left.\frac{\mathrm{d}}{\mathrm{d}t} v(t)\right|_{t=0} = \frac{1}{2}\left.\frac{\mathrm{d}}{\mathrm{d}t} \det g_{ij}(t)\right|_{t=0} = \frac{1}{2}\sum_{i=1}^k \left.\frac{\mathrm{d}}{\mathrm{d}t} g_{ii}(t)\right|_{t=0}$$

$$= \sum_{i=1}^k \langle \nabla_{\partial_t F}\partial_i F, \partial_i F\rangle = \sum_{i=1}^k \langle \nabla_{\partial_i F}\partial_t F, \partial_i F\rangle = \mathrm{div}_\Sigma X,$$

which gives the desired equality.

In order to prove (2.2), decompose $X = X^\perp + X^\top$, which are its normal and tangential components respectively. For an orthonormal tangent frame $\{e_1, \dots, e_k\}$, we have

$$\mathrm{div}_\Sigma X^\perp = \sum_{i=1}^k \langle D_{e_i} X^\perp, e_i\rangle = -\sum_{i=1}^k \langle X^\perp, D_{e_i} e_i\rangle$$

$$= -\sum_{i=1}^k \langle X^\perp, A(e_i, e_i)\rangle = -\langle X^\perp, \vec{H}\rangle = -\langle X, \vec{H}\rangle,$$

where we are using that $\langle X^\top, \vec{H}\rangle = 0$. Since $X^\top$ is tangent to Σ we have

$$\operatorname{div}_\Sigma X^\top = \operatorname{div}_\Sigma^{\text{intrinsic}} X^\top.$$

We get (2.2) by the intrinc divergence theorem, indeed:

$$\begin{aligned}\int_\Sigma \operatorname{div}_\Sigma X \mathrm{d}\operatorname{vol}_\Sigma &= \int_\Sigma \operatorname{div}_\Sigma X^\perp \mathrm{d}\operatorname{vol}_\Sigma + \int_\Sigma \operatorname{div}_\Sigma X^\top \mathrm{d}\operatorname{vol}_\Sigma \\ &= -\int_\Sigma \langle \vec{H}, X\rangle \mathrm{d}\operatorname{vol}_\Sigma + \int_{\partial\Sigma} \langle X, \eta\rangle \mathrm{d}\operatorname{vol}_{\partial\Sigma},\end{aligned}$$

where one uses $\langle X^\perp, \eta\rangle = 0$. □

Definition 2.4
A submanifold $\Sigma \subset M$ is *minimal* if its mean curvature vector $\vec{H}$ vanishes identically. Equivalently, Σ is a critical point of the area functional for every compactly supported variation fixing $\partial\Sigma$.

Remark 2.3
Area minimizers are minimal, but minimal submanifolds need not globally minimize area.

2.2 Examples in $\mathbb{R}^3$

- **Plane:** $\Sigma = \{z = 0\} \subset \mathbb{R}^3$ has $\vec{H} = 0$.
- **Disk:** $\Sigma = \{(x, y, 0) : x^2 + y^2 \le 1\}$ is minimal with boundary. For compactly supported variations and constant vertical displacement $X = (0, 0, 1)$ the first variation is zero; an outward radial variation $X = (x, y, 0)$ changes its area, owing to the boundary term:

$$\delta\Sigma(X) = 2\pi = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \pi(1 + t)^2.$$

- **Catenoid:** A surface of revolution, for instance $x^2 + y^2 = \cosh^2 z$. Two sufficiently separated coaxial circles cannot be joined by an area-minimizing catenoid; compare with two disks. Discovered by Euler (1744) [11]; studied by Meusnier (1776, with a memoir published in 1785) [23]. Recall $\cosh z = (e^z + e^{-z})/2$.
- **Helicoid:** One parametrization is

$$F(t, s) = (t\cos s, t\sin s, s), \qquad s, t \in \mathbb{R}.$$

  Away from the axis its equation can be expressed locally as $z = \arctan(y/x)$, modulo branches. The helicoid is a complete embedded minimal surface, simply connected, and ruled. Meusnier (1776); Catalan (1842) showed it is the only nonplanar ruled minimal surface [4]. For illustrations of the helicoid, see the gallery by Matthias Weber [34].

2.3 The minimal surface equation for graphs

For a graph $\Sigma^n = \{(x, u(x)) : x \in \Omega \subset \mathbb{R}^n\} \subset \mathbb{R}^{n+1}$, the minimal surface equation is

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0.$$

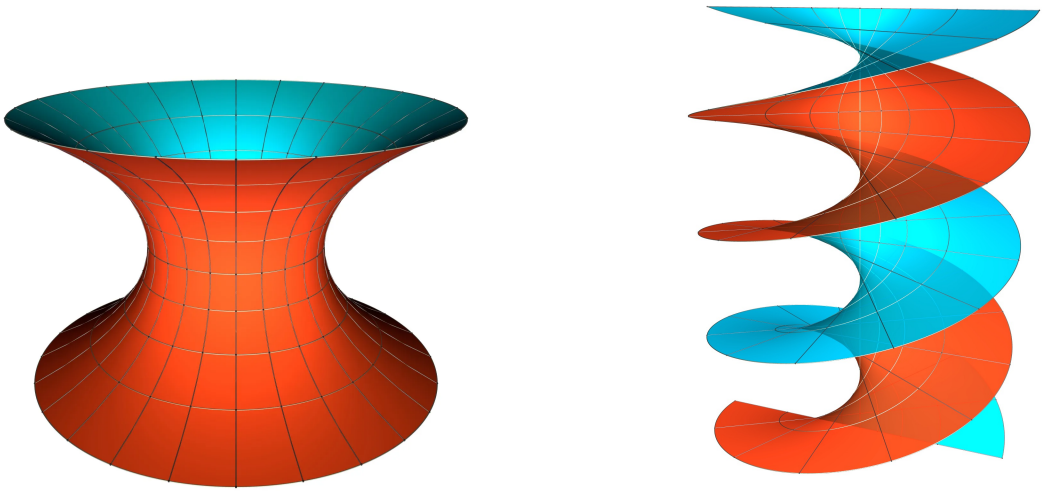


Figure 2.1: Catenoid (left) and helicoid (right) from the minimal surfaces gallery of Matthias Weber [34].

In coordinates this can be written as

$$\Delta u - \frac{\sum_{i,j=1}^{n} u_i u_j u_{ij}}{1+|\nabla u|^2} = 0, \qquad \text{or} \qquad \sum_{i,j=1}^{n} \left(\delta_{ij} - \frac{u_i u_j}{1+|\nabla u|^2}\right) u_{ij} = 0.$$

This is a second-order quasilinear elliptic PDE.

2.4 First consequences of minimality

Proposition 2.5
A submanifold $\Sigma^k \subset \mathbb{R}^n$ is minimal if and only if the restrictions of the Euclidean coordinate functions $x^1, \dots, x^n$ to Σ are harmonic.

Proof. Let $E_j = \nabla x^j$ in $\mathbb{R}^n$ and $\varphi \in C_c^\infty(\Sigma \setminus \partial\Sigma)$. Set $X = \varphi E_j$. Since the Euclidean connection makes E_j parallel in cap 0 mettere definizione trasporto parallelo,

$$\operatorname{div}_\Sigma X = \sum_{i=1}^{k} \langle \nabla_{e_i}(\varphi E_j), e_i \rangle = \langle \nabla_\Sigma \varphi, E_j \rangle.$$

Therefore

$$\begin{aligned} \int_\Sigma \varphi \Delta_\Sigma x^j \,\mathrm{d}\operatorname{vol}_\Sigma &= -\int_\Sigma \langle \nabla_\Sigma \varphi, \nabla_\Sigma x^j \rangle \,\mathrm{d}\operatorname{vol}_\Sigma \\ &= -\int_\Sigma \operatorname{div}_\Sigma(\varphi E_j) \,\mathrm{d}\operatorname{vol}_\Sigma = \int_\Sigma \varphi \langle \vec{H}, E_j \rangle \,\mathrm{d}\operatorname{vol}_\Sigma . \end{aligned}$$

Thus $\Delta_\Sigma \vec{x} = \vec{H}$ and the claim follows. □

Proposition 2.6
Let $\Sigma^k \subset \mathbb{R}^n$ be a compact minimal submanifold with boundary. Then

$$k \operatorname{area}(\Sigma) = \int_{\partial\Sigma} \langle x, \eta \rangle \,\mathrm{d}\operatorname{vol}_{\partial\Sigma} .$$

If $\Sigma \subset B_R(0)$ and $\partial\Sigma \subset \partial B_R(0)$, then

$$\operatorname{area}(\Sigma) \leq \frac{R}{k} \operatorname{area}(\partial\Sigma).$$

Proof. Take $X(x) = x$. In Euclidean space $\nabla_v X = v$, so $\operatorname{div}_\Sigma X = k$. Apply the first variation/divergence formula and $\vec{H} = 0$. □

Remark 2.4
For a minimal $\Sigma^k \subset \mathbb{R}^n$, $\Delta_\Sigma |x|^2 = 2k$.

Proposition 2.7 - Convex hull property
If $\Sigma^k \subset \mathbb{R}^n$ is a compact minimal submanifold with boundary, then

$$\Sigma \subset \operatorname{Conv}(\partial\Sigma).$$

Proof. For every fixed unit vector v, the function $u(x) = \langle v, x \rangle$ is harmonic on Σ and attains its maximum on $\partial\Sigma$ by the maximum principle. Hence any half-space containing $\partial\Sigma$ contains Σ. Intersecting these half-spaces proves the claim. □

Corollary 2.8
There are no compact minimal submanifolds of $\mathbb{R}^n$ without boundary (of positive dimension).

Corollary 2.9
Let $\Sigma^2 \subset \mathbb{R}^n$ be a compact minimal surface with boundary, homeomorphic to a disk. Let $K \subset \mathbb{R}^n$ be compact and convex with $K \cap \partial\Sigma = \emptyset$. Then $K \cap \Sigma$ is simply connected.

Sketch. By contradiction, suppose that a closed curve $\gamma \subset K \cap \Sigma$ does not bound a disk in $K \cap \Sigma$. Since Σ is disk-like, γ bounds a disk $\Gamma \subset \Sigma$. By the convex hull property, $\Gamma \subset \operatorname{Conv}(\gamma) \subset K$, a contradiction. □

Corollary 2.10
Let $\Sigma^k \subset \mathbb{R}^n$ be a compact minimal submanifold and K compact convex, with $K \cap \partial\Sigma = \emptyset$. Then the inclusion $\Sigma \cap K \hookrightarrow \Sigma$ induces an injection on the $(k-1)$-st homology group.

2.5 Monotonicity formula

Let $\Sigma^k \subset \mathbb{R}^n$ be a minimal submanifold and set $\omega_k = |B_1^k(0)|$. We denote by $B_r(x_0)$ the Euclidean ball in $\mathbb{R}^n$, and by $B_r^\Sigma(o)$ the intrinsic geodesic ball in Σ.

Theorem 2.11 - Monotonicity formula(i) *(Extrinsic version) For $x_0 \in \mathbb{R}^n$ and $0 < s < r < \operatorname{dist}(x_0, \partial\Sigma)$, assuming the intersections have finite volume, we have*

$$\frac{|\Sigma \cap B_r(x_0)|}{r^k} - \frac{|\Sigma \cap B_s(x_0)|}{s^k} = \int_{\Sigma \cap (B_r(x_0) \setminus B_s(x_0))} \frac{|(x - x_0)^\perp|^2}{|x - x_0|^{k+2}} \, \mathrm{d}\operatorname{vol}_\Sigma . \tag{2.3}$$

In particular,

$$\Theta_\Sigma^{\mathrm{ext}}(x_0, r) := \frac{|\Sigma \cap B_r(x_0)|}{\omega_k r^k}$$

is monotone nondecreasing. It is constant in r if and only if Σ is conical with vertex x_0 (in the relevant annulus).

If $x_0 \in \Sigma$ is a regular point, then $\Theta^{\mathrm{ext}}_\Sigma(x_0, r) \geq 1$; equality for some r holds if and only if $\Sigma \cap B_r(x_0)$ is a flat k-disk, under the usual connectedness and no-boundary assumptions.

(ii) *(Intrinsic version) For $x_0 \in \Sigma$ and $0 < s < r < \operatorname{dist}_\Sigma(x_0, \partial\Sigma)$, write $r_\Sigma(x) = \operatorname{dist}_\Sigma(x_0, x)$ and choose Euclidean coordinates so that x_0 is mapped to 0. Then*

$$\frac{|B^\Sigma_r(x_0)|}{r^k} - \frac{|B^\Sigma_s(x_0)|}{s^k} = \int_{B^\Sigma_r(x_0)\setminus B^\Sigma_s(x_0)} \frac{1}{r_\Sigma(x)^{k+1}} \Bigg[\big(r_\Sigma(x) - |x|\big) + \frac{|x - x_0|}{2} \left| \frac{x - x_0}{|x - x_0|} - \nu_{B^\Sigma_{r_\Sigma(x)}(x_0)}(x) \right|^2 \Bigg] \mathrm{d}\,\mathrm{vol}_\Sigma(x). \tag{2.4}$$

Here $\nu_{B^\Sigma_t(x_0)}$ is the outward intrinsic unit normal, defined almost everywhere, and the integrand at $x = 0$ is interpreted by continuity. Thus

$$\Theta^{\mathrm{int}}_\Sigma(x_0, r) := \frac{|B^\Sigma_r(x_0)|}{\omega_k r^k}$$

is also monotone nondecreasing, with $\Theta^{\mathrm{int}}_\Sigma(x_0, r) \geq 1$. The intrinsic formula holds with the standard almost-everywhere interpretation at the cut locus [15].

Proof. For the extrinsic formula, we can assume $x_0 = 0$ and that Σ intersects ∂B_s and ∂B_r transversally. Consider

$$X(x) = \frac{x}{|x|^k}, \qquad DX = \frac{\mathrm{Id}}{|x|^k} - \frac{k\, x \otimes x^\flat}{|x|^{k+2}}.$$

If $\{e_i\}_{i=1}^k$ is an orthonormal basis of $T_x\Sigma$, then

$$\operatorname{div}_\Sigma X = \sum_{i=1}^k \langle D_{e_i} X, e_i \rangle = \frac{k}{|x|^k} - \frac{k|x^\top|^2}{|x|^{k+2}} = \frac{k|x^\perp|^2}{|x|^{k+2}}.$$

Integrating on $\Sigma \cap (B_r \setminus B_s)$ and applying the divergence theorem, together with $\operatorname{div}_\Sigma x = k$, gives (2.3). The general result follows by approximation by regular radii. If the density is constant, $x^\perp = 0$ in the annulus, hence the radial vector field is tangent to Σ and Σ is conical. Finally, if $x_0 \in \Sigma$ is regular, $\lim_{r\to 0} \Theta^{\mathrm{ext}}_\Sigma(x_0, r) = 1$.

For the intrinsic formula, assume again that $x_0 = 0$. Recall that $\Delta_\Sigma |x|^2 = 2k$, so the divergence theorem on a regular intrinsic ball yields

$$k|B^\Sigma_R| = \frac{1}{2} \int_{B^\Sigma_R} \Delta_\Sigma |x|^2 \,\mathrm{d}\,\mathrm{vol}_\Sigma = \int_{\partial B^\Sigma_R} \langle x, \nu_{B^\Sigma_R} \rangle \,\mathrm{d}\,\mathrm{vol}_{\partial B^\Sigma_R}.$$

Since

$$\langle x, \nu_{B^\Sigma_R} \rangle = R - \left[(R - |x|) + \frac{|x|}{2} \left| \frac{x}{|x|} - \nu_{B^\Sigma_R} \right|^2 \right],$$

we get

$$R|\partial B^\Sigma_R| - k|B^\Sigma_R| = \int_{\partial B^\Sigma_R} \left[(R - |x|) + \frac{|x|}{2} \left| \frac{x}{|x|} - \nu_{B^\Sigma_R} \right|^2 \right] \mathrm{d}\,\mathrm{vol}_{\partial B^\Sigma_R}.$$

Now use

$$\frac{\mathrm{d}}{\mathrm{d}R}\left(\frac{|B_R^\Sigma|}{R^k}\right) = \frac{R|\partial B_R^\Sigma| - k|B_R^\Sigma|}{R^{k+1}}$$

and the coarea formula to obtain (2.4). Since $r_\Sigma(x) \geq |x|$, its right-hand side is nonnegative. □

Remark 2.5
The monotonicity formulas also hold for immersed submanifolds, with volume counted with multiplicity. In particular, $\lim_{r\to 0} \Theta_\Sigma^{\mathrm{ext}}(x_0, r)$ is the multiplicity at x_0 (for a smooth immersion). The extrinsic formula extends to weaker notions of minimal submanifolds and is fundamental in regularity theory.

Corollary 2.12 - Florit–Simon [15]
Let $\Sigma^k \subset \mathbb{R}^n$ be a complete, connected, smoothly immersed minimal submanifold, without boundary, and $o \in \Sigma$. Then

$$\lim_{R\to\infty} \Theta_\Sigma^{\mathrm{int}}(o, R) = \lim_{R\to\infty} \Theta_\Sigma^{\mathrm{ext}}(o, R) \in [1, +\infty].$$

In particular, bounded intrinsic area density is equivalent to bounded extrinsic area density, and either bound implies that the immersion is proper.

Remark 2.6
The properness assumption cannot be omitted in statements about bounded complete minimal surfaces. Jorge–Xavier constructed a complete minimal immersion in $\mathbb{R}^3$ between two parallel planes [19], and Nadirashvili constructed a complete minimal immersion in $\mathbb{R}^3$ contained in a ball [27]. Both are nonproper.

Theorem 2.13 - Extended monotonicity
Let $\Sigma^k \subset \mathbb{R}^n$ be a compact minimal submanifold with smooth boundary and $x_0 \in \mathbb{R}^n \backslash \partial\Sigma$. Define the exterior cone over $\partial\Sigma$ with vertex x_0 by

$$E = E(x_0, \partial\Sigma) := \{x_0 + t(x - x_0) : t \geq 1,\ x \in \partial\Sigma\}, \qquad \widetilde{\Sigma} = \Sigma \cup E.$$

The volume of E is counted with multiplicity when needed. Then

$$\Theta_{\widetilde{\Sigma}}^{\mathrm{ext}}(x_0, r) := \frac{|\widetilde{\Sigma} \cap B_r(x_0)|}{\omega_k r^k}$$

is monotone nondecreasing for $r > 0$. Equality on an interval of radii forces the radial vector field to be tangent to $\widetilde{\Sigma}$ in the corresponding annulus, with vanishing boundary contribution along $\partial\Sigma$.

Proof. On the cone E, the position vector $x - x_0$ is tangent, so $\langle x - x_0, \vec{H}_E\rangle = 0$. On $\partial\Sigma$, the outward conormals of Σ and E satisfy

$$\langle x - x_0, \eta_\Sigma + \eta_E\rangle \leq 0.$$

Using the divergence theorem for $X = (x - x_0)/|x - x_0|^k$ on both pieces, the common boundary gives a nonnegative contribution to the variation of the density. The remaining term is the same nonnegative integrand as in (2.3). □

Theorem 2.14 - Fáry–Milnor [12, 24]
Let $\Gamma \subset \mathbb{R}^3$ be a smooth simple closed curve. If

$$\int_\Gamma |\vec{k}| \, \mathrm{d}s \leq 4\pi,$$

where $\vec{k}$ is the curvature vector, then Γ is unknotted.

Sketch. First assume $\int_\Gamma |\vec{k}| \, \mathrm{d}s < 4\pi$. Let Σ be an area-minimizing disk with boundary Γ, provided by the Douglas–Radó solution to the Plateau problem [10, 30]. Interior regularity gives a smooth minimal immersion (and in this setting no interior branch points). It is enough to prove that Σ is embedded.

Suppose by contradiction that two sheets of Σ meet at an interior point $x_0 \in \Sigma$. Then

$$\lim_{r \to 0} \Theta^{\mathrm{ext}}_{\Sigma}(x_0, r) \geq 2.$$

Let $E = E(x_0, \Gamma)$ and $\widetilde{\Sigma} = \Sigma \cup E$. By extended monotonicity,

$$2 \leq \lim_{r \to 0} \Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, r) \leq \lim_{R \to \infty} \Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, R). \tag{2.5}$$

On the other hand, the compact part Σ does not contribute to the density at infinity. By the conical structure,

$$\lim_{R \to \infty} \frac{|E \cap B_R(x_0)|}{\pi R^2} = \lim_{R \to \infty} \frac{|E \cap \partial B_R(x_0)|}{2\pi R}.$$

The annulus $E \cap B_R(x_0)$ is flat away from the vertex. By Gauss–Bonnet, its two boundary components have opposite total geodesic curvature. The outer boundary contributes its length divided by R, and the contribution at Γ is bounded by the total curvature of Γ:

$$\frac{|E \cap \partial B_R(x_0)|}{R} = -\int_\Gamma \langle \vec{k}, \eta_E \rangle \, \mathrm{d}s \leq \int_\Gamma |\vec{k}| \, \mathrm{d}s.$$

Therefore

$$\lim_{R \to \infty} \Theta^{\mathrm{ext}}_{\widetilde{\Sigma}}(x_0, R) \leq \frac{1}{2\pi} \int_\Gamma |\vec{k}| \, \mathrm{d}s < 2,$$

contradicting (2.5). The equality case $\int_\Gamma |\vec{k}| \, \mathrm{d}s = 4\pi$ follows from the rigidity argument in the Fáry–Milnor theorem [12, 24]. □

2.6 The halfspace theorem

Theorem 2.15 - Hoffman–Meeks halfspace theorem [17]
Let $\Sigma^2 \subset \mathbb{R}^3$ be a connected, complete, properly embedded minimal surface. If Σ is contained in a halfspace, then Σ is a plane.

To prove this result, we need a version of the maximum principle for minimal hypersurfaces.

Theorem 2.16 - Strong maximum principle
If $\Sigma_1, \Sigma_2 \subset \mathbb{R}^{n+1}$ are connected minimal hypersurfaces, tangent at an interior point, and one lies locally on one side of the other, then they agree locally. Consequently, if $U \subset \mathbb{R}^{n+1}$ is open, $\Sigma_1 \cap U$ and $\Sigma_2 \cap U$ are connected, $\Sigma_1 \cap U = \partial\Omega$ for an open set

$\Omega \subset U$, and $\Sigma_2 \cap U \subset \overline{\Omega}$, then either the two pieces are disjoint or they agree.

Sketch of the halfspace theorem. We use the family of half-catenoids as barriers. Up to a rigid motion, assume $\Sigma \subset \{z \geq 0\}$ and that $\inf_\Sigma z = 0$. If Σ touches $\{z = 0\}$, the maximum principle implies that it is a plane. Thus we may assume that $\Sigma \cap \{z = 0\} = \varnothing$.

For $r > 0$ and $\varepsilon > 0$, consider the lower half of a catenoid with neck of radius r at height ε:

$$\Gamma_{\varepsilon,r} = \left\{ (x,y,z) \in \mathbb{R}^3 : \begin{array}{c} x^2 + y^2 = r^2 \cosh^2\left(\dfrac{z-\varepsilon}{r}\right), \\ z \leq \varepsilon \end{array} \right\}.$$

Its boundary is the circle $\{x^2 + y^2 = r^2,\ z = \varepsilon\}$. Properness allows us to choose $\varepsilon > 0$ sufficiently small so that Σ is disjoint from $\Gamma_{\varepsilon,1}$ and from its boundary. As $r \downarrow 0$, $\Gamma_{\varepsilon,r}$ converges, away from the vertical axis, to the punctured plane $\{z = \varepsilon\}$, which meets Σ since $\inf_\Sigma z = 0$.

We now vary r between 0 and 1. For $0 < r \leq 1$, the boundaries $\partial\Gamma_{\varepsilon,r}$ remain disjoint from Σ. Moreover, any intersections lie in a fixed compact region: far from the axis the half-catenoids lie below $\{z = 0\}$. At a first value of r for which the two surfaces cease to intersect, there is therefore an interior tangency. This contradicts the maximum principle. The complete argument, including the barrier construction, is in [17]. □

Remark 2.7
The hypothesis of properness is essential. Jorge–Xavier constructed a complete immersed minimal surface between two parallel planes [19].

Remark 2.8
Complete embedded minimal surfaces of finite topology in $\mathbb{R}^3$ are proper. This is a consequence of the work of Colding–Minicozzi [6]. It remains open whether every complete embedded minimal surface of finite genus is proper. Properness is known when the set of ends is countable [21].

The Riemann minimal examples, introduced around 1860, are complete, properly embedded, singly periodic minimal surfaces of genus zero with infinitely many planar ends. They provide examples of much more complicated global geometry than planes, catenoids, and helicoids.

2.7 The Weierstrass representation

We now consider minimal surfaces Σ^2 in $\mathbb{R}^3$. Complex geometry provides a powerful method for producing examples.

Let $F\colon \Sigma \to \mathbb{R}^3$ be smooth. Recall that F is an *immersion* if dF_p is injective for every $p \in \Sigma$, a *branched immersion* if this fails only at isolated points, and an *embedding* if F is an immersion and a homeomorphism onto its image.

Theorem 2.17 - Isothermal coordinates
Every point of a smooth Riemannian surface (Σ^2, g) admits local coordinates (x, y) such that

$$g = \mu^2(dx^2 + dy^2), \qquad \mu > 0.$$

Moreover, x and y are harmonic with respect to g.

Set $z = x + iy$, $\overline{z} = x - iy$, and

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\overline{z}} = \frac{1}{2}(\partial_x + i\partial_y).$$

Since $\Delta_\Sigma = \mu^{-2}(\partial_x^2 + \partial_y^2)$ on functions, a map $F\colon \Sigma \to \mathbb{R}^3$ is harmonic if and only if $\partial_{\overline{z}}\partial_z F = 0$.

Suppose that F is an isothermal immersion and define

$$\Phi = 2\partial_z F = F_x - iF_y = (\phi_1, \phi_2, \phi_3).$$

Conformality is equivalent to

$$\Phi \cdot \Phi = \phi_1^2 + \phi_2^2 + \phi_3^2 = 0, \qquad |\Phi|^2 = 2\mu^2,$$

where the first dot product is complex bilinear and the second norm is Hermitian. Moreover,

$$F \text{ is minimal} \iff \Delta_\Sigma F = 0 \iff \partial_{\overline{z}}\Phi = 0.$$

Thus the components of Φ are holomorphic. Also,

$$dF = \operatorname{Re}(\Phi\, dz), \qquad F(p) = F(p_0) + \operatorname{Re}\int_{p_0}^{p} \Phi\, dz.$$

If Σ is not simply connected, the integral is independent of path precisely when all real periods vanish.

Assume that F is oriented, and let $N\colon \Sigma \to S^2$ be its unit normal, with the orientation induced by F. Denote by $\pi\colon S^2 \to \mathbb{C} \cup \{\infty\}$ stereographic projection from the north pole. The map $g := \pi \circ N$ is the *meromorphic Gauss map*. In terms of Φ it is given by

$$g = \frac{\phi_3}{\phi_1 - i\phi_2} = -\frac{\phi_1 + i\phi_2}{\phi_3},$$

whenever the quotients make sense. It follows that

$$\phi_1 = \frac{1}{2}\left(g^{-1} - g\right)\phi_3, \qquad \phi_2 = \frac{i}{2}\left(g^{-1} + g\right)\phi_3.$$

Writing $dh = \phi_3\, dz$, we obtain the Weierstrass representation

$$\boxed{F = \operatorname{Re}\int \left(\frac{1}{2}\left(g^{-1} - g\right), \frac{i}{2}\left(g^{-1} + g\right), 1\right) dh.}$$

Here dh is a holomorphic 1-form, and the apparent singularities in the expression must cancel so that the vector-valued 1-form is holomorphic. The induced metric is

$$ds^2 = \frac{1}{4}\left(|g| + |g|^{-1}\right)^2 |dh|^2.$$

At zeros and poles of g the expression is interpreted by continuity.

Proposition 2.18 - Weierstrass representation
Let Σ be a Riemann surface, g a meromorphic function, and dh a holomorphic 1-form such that

$$\Phi\, dz = \left(\frac{1}{2}(g^{-1} - g), \frac{i}{2}(g^{-1} + g), 1\right) dh$$

is holomorphic and has no common zeros in its three components. If all its periods are purely imaginary, then

$$F(p) = \operatorname{Re}\int_{p_0}^{p} \Phi\, dz$$

defines a conformal minimal immersion $F\colon \Sigma \to \mathbb{R}^3$. If the components have isolated

common zeros, the same formula defines a branched minimal immersion. On a simply connected Σ, the period condition is automatic.

The Gauss map can be recovered from g by

$$N = \frac{1}{1+|g|^2}(2\operatorname{Re} g, 2\operatorname{Im} g, |g|^2 - 1).$$

Indeed, the right-hand side has norm one and is orthogonal to both $\operatorname{Re}\Phi$ and $\operatorname{Im}\Phi$.

Example 2.9 - Classical Weierstrass data (i) The *catenoid*: $\Sigma = \mathbb{C}\setminus\{0\}$,

$$g(z) = z, \qquad dh = \frac{dz}{z}.$$

(ii) The *helicoid*: $\Sigma = \mathbb{C}$,

$$g(z) = e^{iz}, \qquad dh = dz.$$

(iii) *Enneper's surface* (1864): $\Sigma = \mathbb{C}$,

$$g(z) = z, \qquad dh = z\,dz.$$

Up to a translation and a dilation, a parametrization in real coordinates $z = s + it$ is

$$F(s,t) = \left(s - \frac{s^3}{3} + st^2, -t + \frac{t^3}{3} - s^2 t,\ s^2 - t^2\right).$$

The Weierstrass representation is especially useful for constructing complete minimal surfaces. However, checking the period condition, completeness, and embeddedness can be difficult.

For instance, Costa [8] constructed a complete minimal immersion in $\mathbb{R}^3$ of genus one with three embedded ends, using elliptic functions. Hoffman–Meeks [18] later proved that this surface is embedded. For many years, the plane, catenoid, and helicoid were the only known complete embedded minimal surfaces of finite topology in $\mathbb{R}^3$.

2.8 The Bernstein theorem

Theorem 2.19 - Bernstein [2]
Let $u\colon \mathbb{R}^2 \to \mathbb{R}$ be an entire smooth solution of the minimal surface equation

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = 0.$$

Then $u(x,y) = ax + by + c$ for some constants $a, b, c \in \mathbb{R}$. Equivalently, an entire minimal graph in $\mathbb{R}^3$ is a plane.

We will use the following two lemmas.

Lemma 2.20 - Curvature estimate
Let $\Sigma^2 \subset \mathbb{R}^3$ be an entire minimal graph. There exists a constant $C > 0$ such that, for every compactly supported Lipschitz function η on Σ,

$$\int_\Sigma \eta^2 |A|^2 \,\mathrm{d}\mu \le C \int_\Sigma |\nabla^\Sigma \eta|^2 \,\mathrm{d}\mu.$$

Proof. Let $N\colon \Sigma \to S^2$ be the upward pointing Gauss map. Since Σ is a graph, its image lies in the open upper hemisphere $S^2_+ = \{v_3 > 0\}$. There exists a smooth 1-form α on the closed upper hemisphere with

$$d\alpha = \mathrm{d}\,\mathrm{vol}_{S^2}, \qquad |\alpha| \le C_0.$$

Since Σ is minimal, $H = 0$ and its principal curvatures satisfy $\lambda_1 = -\lambda_2$. Consequently,

$$|A|^2 = \lambda_1^2 + \lambda_2^2 = -2\lambda_1\lambda_2 = -2K, \qquad N^*(\mathrm{d}\,\mathrm{vol}_{S^2}) = K\,\mathrm{d}\mu.$$

In particular,

$$\begin{aligned}\int_\Sigma \eta^2 |A|^2\,\mathrm{d}\mu &= -2\int_\Sigma \eta^2\, d(N^*\alpha)\\ &= 4\int_\Sigma \eta\, d\eta \wedge N^*\alpha\\ &\le 4C_0 \int_\Sigma \eta |\nabla^\Sigma \eta||A|\,\mathrm{d}\mu.\end{aligned}$$

The conclusion follows from Cauchy–Schwarz (and approximation for Lipschitz cutoffs). □

Lemma 2.21 - Quadratic area bound
Let $\Sigma^2 \subset \mathbb{R}^3$ be an entire minimal graph. Then for every $r > 0$ and every $x_0 \in \mathbb{R}^3$,

$$|\Sigma \cap B_r(x_0)| \le 2\pi r^2.$$

Proof. Extend the upward unit normal N to $\mathbb{R}^3$ by keeping it constant along vertical lines. Consider the 2-form

$$\omega(X, Y) = \det(X, Y, N).$$

Since u solves the minimal surface equation, $\operatorname{div} N = 0$, and hence $d\omega = 0$. Moreover, $|\omega(X, Y)| \le 1$ for orthonormal X, Y, with equality on the oriented tangent planes of Σ. Thus ω calibrates Σ, and Σ is area-minimizing.

For almost every r, $\Sigma \cap \partial B_r(x_0)$ is a smooth union of closed curves. The graph separates $B_r(x_0)$ into an upper and a lower region; the corresponding pieces of $\partial B_r(x_0)$ have areas adding up to $4\pi r^2$. By the calibration and Stokes' theorem, $\Sigma \cap B_r(x_0)$ has area no larger than either of these two spherical pieces. Therefore

$$|\Sigma \cap B_r(x_0)| \le \frac{1}{2}|\partial B_r(x_0)| = 2\pi r^2.$$

The conclusion for every r follows by approximation. □

Proof of Bernstein's theorem. Fix $R > 2$ and define the radial cutoff $\eta_R\colon \mathbb{R}^3 \to [0, 1]$ by

$$\eta_R(x) = \begin{cases} 1, & |x| \le R,\\ 2 - \dfrac{\log|x|}{\log R}, & R < |x| < R^2,\\ 0, & |x| \ge R^2. \end{cases}$$

Since $|\nabla^\Sigma |x|| \le 1$, we have

$$|\nabla^\Sigma \eta_R| \le \frac{1}{|x| \log R} \quad \text{on } B_{R^2} \setminus B_R.$$

By the curvature estimate and the quadratic area bound,

$$\begin{aligned}\int_{\Sigma\cap B_R} |A|^2 \,\mathrm{d}\mu &\le \int_\Sigma \eta_R^2 |A|^2 \,\mathrm{d}\mu \\ &\le \frac{C}{(\log R)^2} \int_{\Sigma\cap(B_{R^2}\setminus B_R)} \frac{1}{|x|^2}\,\mathrm{d}\mu \\ &\le \frac{C'}{\log R}.\end{aligned}$$

For the last inequality, decompose the annulus into dyadic annuli, or integrate by parts the area function $A(r) = |\Sigma \cap B_r|$, using $A(r) \le 2\pi r^2$. Letting $R \to \infty$ gives $|A| \equiv 0$. Thus Σ is totally geodesic and therefore a plane. □

Remark 2.10 - The Bernstein problem in higher dimensions
A natural question is whether an entire minimal graph $\Sigma^n \subset \mathbb{R}^{n+1}$ must be a hyperplane. The answer is positive for $n \le 7$, by results of Fleming [14], De Giorgi [9], Almgren [1], and Simons [33].

In contrast, Bombieri–De Giorgi–Giusti [3] constructed non-affine entire solutions for $n \ge 8$. The related Simons cone in $\mathbb{R}^8$ is

$$\mathcal{C} = \{x \in \mathbb{R}^8 : x_1^2 + x_2^2 + x_3^2 + x_4^2 = x_5^2 + x_6^2 + x_7^2 + x_8^2\},$$

a singular area-minimizing hypersurface. Thus the dimension threshold for the Bernstein problem is sharp.

2.9 Exercises

Exercise 2.11 - Catenoids
For all $\lambda > 0$, define $C_\lambda \subset \mathbb{R}^3$ to be the following surface of revolution, called *catenoid*,

$$C_\lambda := \left\{(u_1, u_2, z) \in \mathbb{R}^3 \;:\; \lambda\cosh(z/\lambda) = \sqrt{u_1^2 + u_2^2}\right\}.$$

(i) Prove that C_λ is a minimal surface for all $\lambda > 0$.

(ii) Show that there exists a unique value $\lambda > 0$ such that C_λ meets the Euclidean sphere $\partial B(0,1)$ orthogonally. The surface C_λ for this choice of λ is called *critical catenoid*.

(iii) Compute the Gaussian curvature K of the catenoid C_λ, and the total curvature $\int_{C_\lambda} |K| d\sigma$ for all $\lambda > 0$..

Exercise 2.12 - Radially symmetric minimal surfaces
Let $\Omega \subset \mathbb{R}^2$ be a connected open set and take a C^2 function $f\colon \Omega \to \mathbb{R}$. Assume that both Ω and f are radially symmetric, and write $f(x) = w(|x|)$ for some C^2 function $w\colon I \to \mathbb{R}$, where $I \subset \mathbb{R}_+$ is an interval.

(i) Prove that $\mathrm{graph}(f)$ is minimal if and only if

$$rw''(r) + w'(r)(1 + (w'(r))^2) = 0 \quad \text{for all } r \in I. \qquad (\mathrm{MSE}:\ ar)$$

(ii) Show that, if (MSE : ar) holds for $r \in [a,b]$, then

$$\log\left(\frac{w'(r)}{\sqrt{1+(w'(r))^2}}\right) - \log\left(\frac{w'(a)}{\sqrt{1+(w'(a))^2}}\right) = \log\left(\frac{a}{r}\right)$$

for all $r \in [a,b]$.

(iii) Determine all radially symmetric solutions to the minimal surface equation defined on the following domains $U \subset \mathbb{R}^2$:

- an annulus of radii a, b, namely $\Omega = \{x \in \mathbb{R}^2 \ : \ a < |x| < b\}$;
- the complement of a ball of radius a, namely $\Omega = \{x \in \mathbb{R}^2 \ : \ |x| > a\}$;
- the whole Euclidean plane $\Omega = \mathbb{R}^2$.

Exercise 2.13 - Ellipticity of minimal surface equation
Recall that the minimal surface equation for the graph of a function $u \colon \Omega \subset \mathbb{R}^n \to \mathbb{R}$ can be written as

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = \Delta u - \frac{D^2u(\nabla u, \nabla u)}{1+|\nabla u|^2} = \sum_{i,j=1}^{n} a_{ij} D^2_{ij} u = 0,$$

where

$$a_{ij} \coloneqq \delta_{ij} - \frac{D_i u D_j u}{1+|\nabla u|^2}.$$

Prove that $(a_{ij})_{i,j=1,\dots,n}$ is a positive definite matrix with eigenvalues 1 with multiplicity $n-1$ and $(1+|\nabla u|^2)^{-1}$ with multiplicity 1. Conclude that the minimal surface equation is not uniformly elliptic.

Exercise 2.14 - The maximum principle
The goal of this exercise is to prove the *strong maximum principle for minimal hypersurfaces*, Theorem 2.16, in the case of surface in $\mathbb{R}^3$ (i.e., $n = 2$ in the statement).

(i) Show that the theorem follows from this local graphical statement. Let $\Omega \subset \mathbb{R}^2$ be an open, bounded, connected neighborhood of the origin. Suppose that $u_1, u_2 \colon \Omega \to \mathbb{R}$ are smooth solutions of the minimal surface equation such that $u_1(0) = u_2(0)$ and $u_1 \leq u_2$. Then $u_1 = u_2$.

(ii) Let us denote by $F \colon \mathbb{R}^2 \to \mathbb{R}^2$ the map

$$F(X) \coloneqq \frac{X}{(1+|X|^2)^{1/2}}.$$

Prove that, for all $X \in \mathbb{R}^2$, the eigenvalues of the matrix $dF(X)$ are bounded between $(1+|X|^2)^{-3/2}$ and $(1+|X|^2)^{-1/2}$.

(iii) Prove that $\operatorname{div}(A(x)\nabla(u_2 - u_1)) = 0$ where the matrix $A(x)$ is given by

$$A(x) \coloneqq \int_0^1 dF(\nabla u_1 + t(\nabla u_2 - \nabla u_1))dt.$$

(iv) Show that the eigenvalues of $A(x)$ are uniformly bounded (from above and from below) by positive constants only depending on $\max_{i=1,2} \sup_{x\in\Omega} |\nabla u_i(x)|$. Namely, there exists a constant $\mu > 0$ that depends on $\max_{i=1,2} \sup_{x\in\Omega} |\nabla u_i(x)|$ such that $\mu < \lambda_1(x) \leq \lambda_2(x) < 1/\mu$ for all $x \in \Omega$, where $\lambda_1(x), \lambda_2(x)$ are the eigenvalues of $A(x)$.

(v) Conclude that $u_1 = u_2$.

Exercise 2.15 - Graphicality
Let $\Omega \subset \mathbb{R}^2$ be a strictly convex, open, bounded, regular domain and let $\gamma \subset \mathbb{R}^3$ be a simple closed curve which is a C^2 graph over $\partial\Omega$. Moreover let $\Sigma \subset \mathbb{R}^3$ be a compact connected minimal surface with $\partial\Sigma = \gamma$. Prove that Σ is graphical over Ω.

Exercise 2.16 - Constant mean curvature surfaces
Let $\Sigma^2 \subset \mathbb{R}^3$ be a closed embedded surface in $\mathbb{R}^3$.

1. Show that $\Sigma = \partial\Omega$ for some open set $\Omega \subset \mathbb{R}^3$.

2. Prove that Σ is a critical point of the area, with respect to the variations that preserve the volume of Ω if and only if Σ has constant mean curvature (often abbreviated CMC).

Index

Divergence along Σ, 21

Monotonicity formula, 25

radial model, 18

two-sided, 17

Bibliography

[1] Frederick J. Almgren, Jr., *Some interior regularity theorems for minimal surfaces and an extension of Bernstein's theorem*, Ann. of Math. (2) **84** (1966), no. 2, 277–292. MR 0200816

[2] Serge Bernstein, *Sur un théorème de géométrie et ses applications aux équations aux dérivées partielles du type elliptique*, Comm. Soc. Math. Kharkow (2) **15** (1915–1917), 38–45.

[3] Enrico Bombieri, Ennio De Giorgi, and Enrico Giusti, *Minimal cones and the Bernstein problem*, Invent. Math. **7** (1969), 243–268. MR 0250205

[4] Eugène Catalan, *Sur les surfaces réglées dont l'aire est un minimum*, J. Math. Pures Appl. **7** (1842), 203–211.

[5] Otis Chodosh, *Introduction to minimal surfaces*, 2025, URL: https://web.stanford.edu/~ochodosh/Math286-min-surf.pdf. Last visited on 2026/10/25. Lecture notes for MATH 286, Stanford, Winter 2025.

[6] Tobias H. Colding and William P. Minicozzi, II, *The Calabi–Yau conjectures for embedded surfaces*, Ann. of Math. (2) **167** (2008), no. 1, 211–243. MR 2373154

[7] Tobias Holck Colding and William P. Minicozzi, II, *A course in minimal surfaces*, Graduate Studies in Mathematics, vol. 121, American Mathematical Society, Providence, RI, 2011.

[8] Celso J. Costa, *Example of a complete minimal immersion in* $\mathbf{R}^3$ *of genus one and three embedded ends*, Bol. Soc. Brasil. Mat. **15** (1984), no. 1-2, 47–54. MR 0794728

[9] Ennio De Giorgi, *Una estensione del teorema di Bernstein*, Ann. Scuola Norm. Sup. Pisa Cl. Sci. (3) **19** (1965), no. 1, 79–85. MR 0178385

[10] Jesse Douglas, *Solution of the problem of Plateau*, Trans. Amer. Math. Soc. **33** (1931), no. 1, 263–321. MR 1501590

[11] Leonhard Euler, *Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes, sive solutio problematis isoperimetrici latissimo sensu accepti*, Marcum-Michaelem Bousquet et socios, Lausanne and Geneva, 1744.

[12] Istvan Fary, *Sur la courbure totale d'une courbe gauche faisant un nœud*, Bull. Soc. Math. France **77** (1949), 128–138. MR 11,393h

[13] Herbert Federer and Wendell H. Fleming, *Normal and integral currents*, Ann. of Math. (2) **72** (1960), no. 3, 458–520. MR 0123260

[14] Wendell H. Fleming, *On the oriented Plateau problem*, Rend. Circ. Mat. Palermo (2) **11** (1962), 69–90.

[15] Enric Florit-Simon, *Equivalence of intrinsic and extrinsic area bounds for minimal surfaces*, 2026, Preprint, arXiv:2605.06468 [math.DG].

[16] Robert D. Gulliver, II, *Regularity of minimizing surfaces of prescribed mean curvature*, Ann. of Math. (2) **97** (1973), no. 2, 275–305. MR 0317188

[17] D. Hoffman and W. H. Meeks, III, *The strong halfspace theorem for minimal surfaces*, Invent. Math. **101** (1990), no. 2, 373–377.

[18] David Hoffman and William H. Meeks, III, *A complete embedded minimal surface in* $\mathbf{R}^3$ *with genus one and three ends*, J. Differential Geom. **21** (1985), no. 1, 109–127.

[19] Luquesio Petrola de Melo Jorge and Frederico Xavier, *A complete minimal surface in* $\mathbb{R}^3$ *between two parallel planes*, Ann. of Math. (2) **112** (1980), no. 1, 203–206. MR 0584079

[20] Joseph-Louis Lagrange, *Essai d'une nouvelle méthode pour déterminer les maxima et les minima des formules intégrales indéfinies*, Œuvres de Lagrange **1** (1867), 335–362, Reprint of the original memoir published in *Miscellanea Taurinensia*, vol. 2 (1760–1761).

[21] William H. Meeks, III, Joaquín Pérez, and Antonio Ros, *The embedded Calabi–Yau conjecture for finite genus*, Duke Math. J. **170** (2021), no. 13, 2891–2956.

[22] William H. Meeks, III and Shing-Tung Yau, *The classical Plateau problem and the topology of three-dimensional manifolds: the embedding of the solution given by Douglas–Morrey and an analytic proof of Dehn's lemma*, Topology **21** (1982), no. 4, 409–442. MR 0670745

[23] Jean Baptiste Marie Meusnier, *Mémoire sur la courbure des surfaces*, Imprimerie Royale, Paris, 1785, Presented at the Académie des Sciences in 1776.

[24] John W. Milnor, *On the total curvature of knots*, Ann. of Math. (2) **52** (1950), no. 2, 248–257. MR 0037509

[25] Charles B. Morrey, Jr., *The problem of Plateau on a Riemannian manifold*, Ann. of Math. (2) **49** (1948), no. 4, 807–851. MR 0027137

[26] Sumner B. Myers, *Riemannian manifolds with positive mean curvature*, Duke Math. J. **8** (1941), no. 2, 401–404. MR 0004518

[27] Nikolai Nadirashvili, *Hadamard's and Calabi–Yau's conjectures on negatively curved and minimal surfaces*, Invent. Math. **126** (1996), no. 3, 457–465. MR 1419004

[28] Robert Osserman, *A proof of the regularity everywhere of the classical solution to Plateau's problem*, Ann. of Math. (2) **91** (1970), no. 3, 550–569. MR 0266070

[29] Joseph Plateau, *Statique expérimentale et théorique des liquides soumis aux seules forces moléculaires*, Gauthier-Villars, Paris, 1873.

[30] Tibor Radó, *On Plateau's problem*, Ann. of Math. (2) **31** (1930), no. 3, 457–469. MR 1502955

[31] Richard Schoen and Shing-Tung Yau, *Existence of incompressible minimal surfaces and the topology of three dimensional manifolds with non-negative scalar curvature*, Ann. of Math. (2) **110** (1979), no. 1, 127–142. MR 0541332

[32] ———, *On the proof of the positive mass conjecture in general relativity*, Comm. Math. Phys. **65** (1979), no. 1, 45–76. MR 0526976

[33] James Simons, *Minimal varieties in Riemannian manifolds*, Ann. of Math. (2) **88** (1968), 62–105.

[34] Matthias Weber, *Minimal Surfaces — repository*, Online collection of minimal-surface illustrations. https://minimalsurfaces.blog/home/repository/. Accessed October 25, 2026.

[35] Brian White, *Introduction to minimal surface theory*, **22** (2016), 387–438.